flattening and gaussian filtering was applied. The dielectric spectra in Fig. 2a were obtained for constant z_{scan} and constant y, but after acquiring the entire 3D data set $|dC/dz|(x,z,f)$ such as shown in Extended Data Fig. 3d. This set is composed of many $|dC/dz|(x,z)$ images (see Extended Data Fig. 3a), which were obtained at different f by scanning the AFM tip across the channels (along the x-axis) at constant y and approaching the surface in steps. This procedure allowed us to minimize errors in z_{scan} due to vertical drifts. When taking $|dC/dz|(x,z)$ images, the AFM tip was approached from large distances towards the surface (down to ~15 nm). At each step, we acquired dozens of lines, which were then averaged to obtain the corresponding profile (Extended Data Fig. 3b). This notably improved the signal-to-noise ratio. The vertical drift of the AFM tip was compensated by comparing the $|dC/dz|(z)$ curve extracted from the 3D data set with a rapidly measured $|dC/dz|(z)$ curve taken above the hBN spacer region immediately before/after the imaging. By taking small steps near to the surface, we could then reconstruct the dielectric spectra (such as in Fig. 2a of the main text) by using the value measured in the middle of nanochannels at the same z_{scan} (± 1 nm) for each f. In addition to the scanning height z_{scan}, the spectral values depended on geometry/size of our devices and of AFM tip's parameters (Extended Data Fig. 10). All the necessary geometric parameters of the studied devices were measured by taking their topography images, whereas the AFM tip parameters (its radius R and half-angle θ) were determined by fitting $|dC/dz|(z)$ curves taken directly above the graphite bottom layer, as in Ref.[22], following the procedures described in Refs.[57,58] For example, the tip radii were found in the range 50-200 nm, in good agreement with the values specified for the commercial diamond-coated tips.

Numerical modeling and data analysis. The observed dielectric spectra were fitted to 3D finite-element numerical calculations implemented using COMSOL Multiphysics 5.4a (AC/DC electrostatic module). These calculations were based on the electrostatic model previously used in Ref.[22], adapted here to simulate the frequency-dependent force acting on the tip as a function of ε and σ of water if an AC electric field was applied. A schematic of the model is shown in Extended Data Fig. 5a. Following Ref.[22], the AFM tip was modeled as a truncated cone with the half-angle θ terminated with a tangent hemispherical apex of radius R. The values of θ and R used in our simulations were measured for each AFM tip, as discussed above. The AFM cone height H_{cone} was limited to 6 µm and the cantilever was omitted to reduce the computational time. We checked that these approximations had no impact on the simulated results. Nanochannels were simulated as 3 rectangular parallelepipeds of length l = 3 µm (Extended Data Fig. 5a). Their height h, width w and spacing w_s were determined from topography imaging for each device. The water channels' electrical response was modeled with the anisotropic, complex dielectric constant

$$\varepsilon^*_{\perp,\parallel}(\omega) = \varepsilon_0\varepsilon_{\perp,\parallel} - i\frac{\sigma_{\perp,\parallel}}{\omega}, \quad (1)$$

where ω is the angular frequency, ε_0 is the dielectric permittivity of vacuum, $\varepsilon_\perp$ and $\varepsilon_{//}$ are the dielectric constants perpendicular and parallel to the channel, respectively, and $\sigma_\perp$ and $\sigma_{//}$ are the perpendicular and parallel conductivities, respectively. Note that in Eq. (1) $\varepsilon_{\perp,//}$ can contain ionic contributions due to ion-water and ion-ion correlations in addition to water's purely dielectric response[59,60].

The dielectric media surrounding the nanochannels (Extended Data Fig. 5a) was modeled as a rectangular cuboid extending above and below the water channels (length l = 3 µm, width W = 3 µm, and height $H_{top} + h + H$), which reflected the corresponding dimensions of the top, bottom and spacer hBN layers. The anisotropic dielectric constant of hBN was set to its bulk values ($\varepsilon_{\perp hBN} \approx 3$, $\varepsilon_{//hBN} \approx 5$). For each device, we numerically solved the Poisson's equation in the frequency domain and calculated the force acting on the AFM probe and, from that, $|dC/dz|$ by integrating the built-in Maxwell stress tensor on the probe's surface using the same simulations' box size and boundary conditions as in Ref.[22] Note that, for brevity, in all the figures the notion $|dC/dz|$ refers to the relative dielectric contrast, that is, the response relatively to the hBN spacer so that $|dC/dz| \equiv dC(z_{scan},\varepsilon_{\perp,//},\sigma_{\perp,//})/dz - dC(z_{scan}, \varepsilon_{\perp,//hBN},0)/dz$. Accordingly, we computed $|dC/dz|$ over the center of the water channel as a function of f and subtracted the corresponding values for the case of the tip placed over the center of the hBN spacer. Examples of the calculated dielectric spectra for representative geometrical parameters of our devices and water's ε and σ are plotted in Extended Data Fig. 6. The plots closely reproduce the experimental behavior (for details, see "Analysis of local dielectric spectra"). For each device, we extracted water's $\varepsilon_{//}$ and $\sigma_{//}$ by fitting the calculated dispersions to the experimental one. Note that $\varepsilon_{//}$ and $\sigma_{//}$ were the only unknowns in our analysis, as all the other parameters needed for the simulations were determined experimentally. $\varepsilon_\perp$ was set to the values measured in Ref.[22] and $\sigma_\perp$ was set to the measured value for our bulk water (σ_{bulk} = 2 x 10^{-4} S·m^{-1}). We also found that the extracted results for $\varepsilon_{//}$ and $\sigma_{//}$ depended little on exact values of $\varepsilon_\perp$ or $\sigma_\perp$, showing that our experimental geometry was rather insensitive to water's out-of-plane characteristics, and the influence was negligible for our smallest channels (Extended Data Figs. 6e,f and 8e,f). This can be understood as being due to the fact that the nanochannel's impedance in the out-of-plane direction is much smaller than the impedance in series due to the hBN layers and AFM tip-surface capacitances, whereas the nanochannel's impedance in the in-plane direction is much larger (see "Electrical modeling" below).

It is important to note that the thickness H of the hBN bottom layer determines whether the measurements are mostly sensitive to $\varepsilon_{//}$ or $\varepsilon_\perp$. Extended Data Fig. 7 illustrates this for the case of channel thickness h = 5 nm at high f, beyond the conductivity relaxation regime (see "Analysis of local dielectric spectra") and for a large AFM tip radius (100 nm) as used in our experiments. When the probed channel is in immediate proximity to a metallic surface (Extended Data Fig. 7a), as in Ref.[22] , the $|dC/dz|$ signal is independent of $\varepsilon_{//}$ and, consequently, the extracted ε represents the out-of-plane component, $\varepsilon_\perp$. Furthermore, measurements are sensitive only to relatively small values of $\varepsilon_\perp$ (up to ~20 for h = 5 nm), as the signal saturates with further increases in $\varepsilon_\perp$ (for larger channel thicknesses, the sensitivity extends to increasingly larger values of $\varepsilon_\perp$, as shown in Ref.[22]). Conversely, when the channel is on a thick hBN layer (H = 200 nm; Extended Data Fig. 7b), the $|dC/dz|$ signal significantly increases for $\varepsilon_{//}$ > 20, becoming dominated by the in-plane component. Therefore, for relatively large values of $\varepsilon_{//}$ ranging from ~80 to ~1,000, as measured in this work, the signal at such H shows little dependence on $\varepsilon_\perp$. Extended Data Fig. 7c shows how the signal evolves as a function of H. In these simulations, we increased the cone height and cantilever length of the AFM tip to their nominal values (H_{cone} = 12 µm and $L_{cantilever}$ = 20 µm). This allowed us to include electrostatic contributions from longer-range components of the AFM probe[61], which we found to be relatively small but still non-negligible for very thick bottom hBN (H > 500 nm). Results show that the $|dC/dz|$ signal and its sensitivity to $\varepsilon_{//}$ peaks between 50 and 500 nm (Extended Data Fig. 7c). For thinner hBN layers, the $\varepsilon_{//}$ contribution is relatively small or negligible, and the signal is dominated by $\varepsilon_\perp$. For hBN thicker than 500 nm, the $|dC/dz|$ signal decreases becoming less sensitive to the probed local electrical properties, consistent with previous results[57]. This is expected because the AFM tip is moved far away from the bottom electrode. In this work we used bottom hBN layers of thickness up to 200 nm. This limitation was dictated by constraints in fabrication of our devices, because thicker hBN crystals were stiffer and provided poor adhesion, leading to delamination upon filling the channels with water.

Analytical modeling. Because of the nonuniform distribution of electric field across nanochannels and the complex geometry of AFM probes, the electrostatic problem cannot be solved exactly using analytical models. Hence, we used the 3D numerical simulations described above. Nonetheless, it is rather informative to also provide an analytical approximation that would substantiate our numerical results and could be used for semi-quantitative estimates. To this end, we employed the point-charge model where the AFM tip was replaced by a point charge Q positioned at distance $z = z_{scan} + R$ from the top hBN surface (Extended Data Fig. 8a). The devices were modeled as a stack of different slabs, infinite in the in-plane direction. The slabs corresponding to the hBN layers were positioned at $0 < z < -H_{top}$ and $-(H_{top} + h) < z < -(H_{top} + h + H)$ and modeled as insulators with zero conductivity. The water layer at $-H_{top} < z < -(H_{top} + h)$ was described by anisotropic dielectric constants $\varepsilon_{//,\perp}$ and conductivities $\sigma_{//,\perp}$. The boundary condition of zero voltage was imposed at $z = -(H_{top} + h + H)$ to model the highly conducting ground electrode. By leveraging the in-plane translational invariance, the Laplace equation for the Fourier transform electrostatic potential in the in-plane direction $\phi_q(z)$ becomes

$$q^2\varepsilon_\parallel(z)\phi_q(z) - \partial_z\left[\varepsilon_\perp(z)\partial_z\phi_q(z)\right] = Q_q\delta(z - z_{scan} - R) \quad (2)$$

where $\boldsymbol{q} = (q_x, q_y)$ is the wave vector in the in-plane direction, and the position-dependent dielectric constants are denoted as $\varepsilon_{//}(z)$ and $\varepsilon_\perp(z)$, respectively. The dielectric constant of the hBN slabs was, for simplicity, assumed isotropic with $\varepsilon_{//}(z) = \varepsilon_\perp(z) = \varepsilon_{hBN} = 4$. The Laplace equation was solved within each slab using exponential functions. The solutions were

matched at each interface by imposing the following boundary conditions: continuity of $\phi_q(z)$ and

$$\varepsilon_\perp(z+\eta)\partial_z\phi_q(z+\eta) - \varepsilon_\perp(z-\eta)\partial_z\phi_q(z-\eta) = 0 \qquad (3)$$

where η is an infinitesimal constant. A similar relation holds at the point-charge location, $z_{scan}+R$, but the right-hand side in (3) becomes equal to Q. The potential in real space $\phi(r,z)$ was then obtained by performing the Fourier transform of $\phi_q(z)$ in the in-plane direction. The capacitance was estimated as $C = Q/\phi(0,z_{scan})$, and from the latter the capacitance gradient $|dC/dz|$ was calculated. Examples of the calculated dielectric spectra for representative parameters of our devices and water's ε and σ are given in Extended Data Fig. 8, where again for brevity $|dC/dz|$ indicates its value relatively to the case of the heterostructure fully made of hBN, $|dC/dz| = dC(z_{scan},\varepsilon_{\perp,//},\sigma_{\perp,//})/dz - dC(z_{scan},\varepsilon_{hBN},0)/dz$. The calculated spectra show that the model captures all the main features of the observed behavior as a function of various parameters and agrees well with our numerical simulations in Extended Data Fig. 6. Note however that, because the analytical model assumes an infinite water layer and does not account for a finite w, the transition between low and high f plateaus becomes somewhat less pronounced (Extended Data Fig. 8d). The analytical results agree with the numerical simulations only for w much larger than the tip radius R (Extended Data Fig. 6d). Accordingly, if only the analytical model were used to fit the reported experimental data, this would result in systematic underestimation of both $\varepsilon_{//}$ and $\sigma_{//}$. In particular, the values extracted for quasi-2D water layers ($h \lesssim 2$ nm) would be underestimated by a factor of about 5, although all the trends with changing h would remain correct.

Analysis of local dielectric spectra. The observed dispersion came from the DC conductivity of water, not from its dipolar orientational (Debye) relaxation[62]. The latter occurs only at $f \approx 10$ GHz in bulk water[63], well above the frequencies considered in our report. The reason for a DC conductivity contribution into our spectra can be understood by noticing that we measured the modulus of dC/dz, that is, the modulus of the complex dielectric constant $|\varepsilon^*_{\perp,\parallel}(\omega)|$. As a result, the dielectric response is determined by both ε and σ and, depending on frequency, either the first or second term in Eq. (1) becomes dominant. At sufficiently low f, the conductivity term always dominates $|\varepsilon^*_{\perp,\parallel}(\omega)|$, similar to the case of macroscale measurements using standard broadband dielectric spectroscopy[64]. To this end, it is useful to recall that at low frequencies the spectrum of deionized water at macroscale is known[63] to be dominated by σ_{bulk}. Therefore, the spectrum is effectively divided into two regions separated by the conductivity relaxation frequency, $f_{r,bulk}$, at which the behavior changes. In the conduction-dominated region ($f < f_{r,bulk}$), the dielectric response decreases with increasing f, while for $f > f_{r,bulk}$ it is constant and depends only on ε_{bulk}. The value of $f_{r,bulk}$ is given by[64]

$$f_{r,bulk} = \frac{\sigma_{bulk}}{2\pi\varepsilon_0\varepsilon_{bulk}} \qquad (4)$$

which follows from Eq. (1) if we use $|\varepsilon_{\perp,\parallel}| = \varepsilon_{bulk}$ and $|\sigma_{\perp,\parallel}| = \sigma_{bulk}$. Note that this frequency is analogous to the cutoff frequency for systems that can be modeled by a simple RC circuit (see "Electrical modeling"). For the bulk water used in our experiments ($\varepsilon_{bulk} \approx 80$, $\sigma_{bulk} \approx 2\cdot10^{-4}$ S·m^{-1}), Eq. (4) yields ~ 45 kHz. This value agrees well with $f_{r,bulk}$ obtained from both numerical and analytical modeling discussed above for the specific experimental geometry. Indeed, the analyses shown in Extended Data Figs. 6b and 8b yielded the purely dielectric behavior characterized by high-f plateaus starting at $f \gtrsim 45$ kHz for all our nanochannels, irrespective of their height h. The additional plateau found in our simulations at low f reflects the presence of non-lossy dielectrics (air gap between the AFM tip and the device; the top and bottom hBN layers). These dielectrics can be represented as additional capacitances in series to water's contribution ("Electrical modeling"). Accordingly, for f below the cut-off frequency $f_{c,bulk}$, the modeled response becomes purely dielectric, reflecting the series capacitances. For $f_{c,bulk} < f < f_{r,bulk}$, the response is dominated by water's σ whereas above the relaxation frequency $f_{r,bulk}$ it is again purely dielectric but now dominated by water's ε. For other relevant values of water's σ and ε, the spectra exhibit similar behavior (Extended Data Figs. 6e,f and 8e,f): there is a low-f plateau up to the cut-off frequency f_c, above which the response decreases with increasing f, and the high-f plateau develops above the conductivity relaxation frequency f_r.

The onset of the low-f plateau with decreasing f is determined by the water's in-plane conductivity $\sigma_{//}$ whereas no information can be inferred about $\sigma_\perp$ from our experimental data because the measurement geometry is insensitive to the latter conductivity. This can be seen in Extended Data Figs. 6e and 8e where the low-f plateau shifts in frequency with varying $\sigma_{//}$, but not $\sigma_\perp$, and completely disappears if $\sigma_{//} = 0$ and $\sigma_\perp \neq 0$. Qualitatively, the stronger dielectric response at low f reflects the fact that the electric potential drops mostly between the AFM tip and the conductive water layer. We used our numerical simulations to illustrate this effect (Extended Data Fig. 5). At low f and $\sigma_{//} \neq 0$, the electric potential drops almost entirely across the air and the top hBN layer with little voltage drop left below the water layer (Extended Data Fig. 5e). Also, the potential distribution extends along the length of nanochannel for a few micrometers (Extended Data Fig. 5c). In contrast, at high $f > f_r$, the potential drop occurs across the entire thickness of our devices, quite similar to the case where the AFM tip is placed above hBN spacers (Extended Data Fig. 5e). The potential drop at high f also extends nearly equally in all the lateral directions around the tip apex (Extended Data Fig. 5d), similar to the case of Extended Data Fig. 5b for non-conducting water layer ($\sigma_{//} = 0$). The effective screening by water's conductivity along the channel length explains the observed large low-f response. Note that the absolute value of the low-f response is independent of $\sigma_{//}$ but depends on geometric parameters, in particular the channel width w, the tip radius R (Extended Data Fig. 6c) and the bottom-layer thickness' H (Extended Data Fig. 7c and Extended Data Fig. 8c). This makes simulations essential for accurate evaluation of the magnitude of changes on the dispersion curves.

While the value of $\sigma_{//}$ does not affect the low-f response, it controls the cut-off frequency f_c, which shifts to higher f proportionally to $\sigma_{//}$ (Extended Data Figs. 6e and 8e) but independently of $\varepsilon_{//}$ (Extended Data Figs. 6f and 8f). This behavior can be described by

$$f_c = \alpha\frac{\sigma_{//}}{2\pi\varepsilon_0}, \qquad (5)$$

where α is the geometrical parameter that can be estimated analytically ("Electrical modeling"). For more accurate results, α can also be obtained using numerical simulations (Extended Data Fig. 10). They yielded $\alpha \approx 2.8 \times 10^{-3}$, 6.2×10^{-4} and 9.2×10^{-5} for our three representative devices discussed in the main text, which had $h \approx 30$, 5 and 1.5 nm, respectively. These values of α suggest that f_c should shift relatively little (from ~10 kHz to ~500 Hz) if water filling the channels exhibited the bulk properties (Extended Data Figs. 6b,c and 8b,c). This shift is much smaller than that observed experimentally and, in fact, occurs in the opposite direction, towards lower f for smaller h, in stark contrast to the experimental behavior (Fig. 2a of the main text). This reiterates the fact that the observed increase in f_c with decreasing h cannot be associated with changes in geometry but comes from a huge increase in water's $\sigma_{//}$ for stronger confinement.

At high f, beyond the conductivity relaxation regime, water behaves as a purely dielectric media and, accordingly, the relative height of the high-f plateau is no longer dependent of $\sigma_{//}$ but depends on water's ε, the channel's geometry (in particular its height h), and the AFM tip radius (Extended Data Fig. 6c). The plateau's height allows us to extract $\varepsilon_{//}$ using numerical simulations. Importantly, for small channels and large in-plane dielectric constant ($\varepsilon_{//} \geq 80$), the contribution of $\varepsilon_\perp$ becomes negligible (Extended Data Figs. 6f and 8f). Furthermore, we find that f_r shifts to higher frequencies proportionally to $\sigma_{//}$ and inversely proportionally to $\varepsilon_{//}$, and is given by

$$f_r = \frac{\sigma_{//}}{2\pi\varepsilon_0\varepsilon_{//}}, \qquad (6)$$

The equation is independent of the device geometry and presents an equivalent of Eq. (4) valid at macroscale. Therefore, once $\sigma_{//}$ is known, $\varepsilon_{//}$ can be directly obtained from f_r using Eq. (6), provided f_r is well separated from f_c as in the case of our small channels. In addition, if α is also known, both $\sigma_{//}$ and $\varepsilon_{//}$ could readily be estimated from the two characteristic frequencies f_c and f_r without the need of simulations, simply using the equations

$$\sigma_{//} = \frac{2\pi\varepsilon_0 f_c}{\alpha}, \qquad (7)$$

$$\varepsilon_{//} = \frac{f_c}{\alpha f_r}, \qquad (8)$$

which follow directly from Eqs. (5-6). Note that the above considerations and Eqs. (5-8) can also be helpful for the analysis of dielectric spectra obtained using other scanning probe approaches[45], including those that probe higher derivatives of $|dC/dz|$ and scanning impedance/microwave microscopy that directly probes the local impedance.

Electrical modeling. It is instructive to use an equivalent impedance circuit to describe the observed dielectric dispersion. However, because of long-range contributions from various AFM cantilever components and complex geometry of our nanochannel devices that result in a nonuniform distribution of the electric field, an equivalent circuit should be so complicated that it is unrealistic to describe our spectra quantitatively. Below, we provide a simplified model that aims to elucidate the physics behind and support our numerical results (Extended Data Fig. 9). To this end, the AFM tip-nanochannel interaction can be modeled by the capacitance, C_{tip}, that for simplicity accounts for both tip-air and top-hBN-layer capacitances and, to a first approximation, can be calculated as $C_{tip} = 2\pi\varepsilon_0 R ln[1+R(1-sin\theta)/(z_{scan}+H_{top}/\varepsilon_{\perp hBN})]$ using the formula described in Ref.[58] We neglect the stray capacitances associated with the AFM cantilever and consider only the tip apex capacitance that is expected to provide the dominant contribution. As for the water-filled nanochannel, we consider it as a distributed RC network shown in Extended Data Fig. 9a. It consists of two elementary RC circuits describing in-plane and out-of-plane impedances $Z_{//}(\omega) = R_{//}/(1+i\omega R_{//}C_{//})$ and $Z_{\perp}(\omega) = R_{\perp}/(1+i\omega R_{\perp}C_{\perp})$, respectively. $R_{\perp,//}$ and $C_{\perp,//}$ can be estimated as $C_{//} = \varepsilon_0\varepsilon_{//}wh/\Delta l$, $R_{//} = \Delta l/(\sigma_{//}wh)$ for the in-plane direction, and $C_{\perp} = \varepsilon_0\varepsilon_{\perp}w\Delta l/h$ and $R_{\perp} = h/(\sigma_{\perp}w\Delta l)$ for the out-of-plane one, where Δl is the length of the circuit element along the channel. We also considered another capacitance C_b in series to $Z_{\perp}$, to model the effect of the bottom hBN layer between the water channel and the ground. Plugging in the experimental values relevant to our devices, one can readily find that $Z_{\perp}(\omega) << 1/(\omega C_b)$ for all frequencies, meaning that the contribution of $Z_{\perp}(\omega)$ is negligible in our experiments, in agreement with the above numerical and analytical calculations. The distributed network can then be simplified further and described by the electrical circuit shown in Extended Data Fig. 9b, where the water impedance is modeled by a single RC unit in the in-plane direction, that is, $Z_{ch}(\omega) = R_{//}/(1+i\omega R_{//}C_{//})$, where $C_{//} = \varepsilon_0\varepsilon_{//}wh/l^*$, $R_{//} = l^*/(\sigma_{//}wh)$ and l^* is the effective length of the nanochannel contributing to electrostatic interactions with the AFM tip. With reference to Extended Data Fig. 5c, l^* notably exceeds the tip diameter and, without loss of generality, can be assumed to be of the order of a few μm. The total equivalent impedance between the AFM tip and ground is then given by

$$Z(\omega) = \frac{1+i\omega R_{//}(C_{//}+C_{geom})}{i\omega C_{geom}(1+i\omega R_{//}C_{//})}, \quad (9)$$

where $C_{geom} = C_{tip}C_b/(C_{tip}+C_b)$ is the capacitance that depends on geometric parameters but not on water's electrical properties. Extended Data Fig. 9c shows $|Z|$ as a function of f for the three representative devices. The effective capacitance of the modeled circuit is given by $1/\omega Z(\omega)$ and displays the same qualitative dependence on f, σ and ε of the capacitance gradient, $|dC/dz|$. Extended Data Fig. 9d plots $|1/\omega Z(\omega)|$ that indeed exhibits both low- and high-f plateaus characterized by frequencies, f_c and f_r, in good agreement with the experiment and numerical simulations. Despite its simplicity, the electrical model also reproduces well the changes in the high-f plateau with varying water's $\varepsilon_{//}$ (Extended Data Fig. 9f) and changes in f_c with varying $\sigma_{//}$ (Extended Data Fig. 9e).

Using this model, we can also corroborate the expressions for f_c and f_r given by Eqs. (5-6). Indeed, the pole of Eq. (9) yields the relaxation frequency as $f_r = 1/(2\pi R_{//}C_{//})$ and, plugging in $R_{//}$ and $C_{//}$ in terms of $\sigma_{//}$ and $\varepsilon_{//}$, results in Eq. (6). The zero of Eq. (9) yields the cut-off frequency so that $f_c \cong 1/(2\pi R_{//}C_{geom})$ where we take into account that C_{geom} is larger than $C_{//}$. Accordingly, f_c is proportional to $\sigma_{//} \propto 1/R_{//}$ and depends on the measurement geometry (through C_{tip} and C_b) but is independent of $\varepsilon_{//}$, in agreement with our numerical simulations. Expressing C_{tip}, C_b and $R_{//}$ in terms of geometric and electrical parameters as defined above and taking $C_b = \varepsilon_0\varepsilon_{hBN}wl^*/H$, we obtain Eq.(5) where, in the case of our geometry, the geometric factor α can be approximated to $\alpha \cong (h\cdot w/l^*)\cdot[1/(2\pi R)+H/(\varepsilon_{hBN}wl^*)]$. Using the specific parameters for our representative devices ($h \approx 30$, 5 and 1.5 nm) and the effective channel length $l^* = 3$ μm, this yields α of about ~ 4 x 10^{-3}, 1 x 10^{-3} and 2 x 10^{-4}, respectively, in reasonable agreement with the numerically simulated values of α. Thus, Eqs. (7-8) with α calculated analytically, as obtained from the simplified electrical model shown in Extended Data Fig. 9b, also allow for semi-quantitative estimates of $\sigma_{//}$ and $\varepsilon_{//}$, which are found to differ only by a factor of < 2 from the values extracted through our more quantitative, numerical analysis.

Data availability statement All relevant data are available from the corresponding authors and at https://zenodo.org/record/12809983

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